

Chapter 5

Object Orientation the new software paradigm

Outline

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Chapter II

Object Orientation the new software paradigm

1. The potential benefits of object orientation
2. The potential drawbacks of object orientation
3. Object standards
4. The object orientation process and models



Object Orientation the new software paradigm

Object-Oriented paradigm

- Involves designing of objects, class and the relationship between class.
 - Class
 - generalized description of an object.
 - A blue print or template which consists of data and operation
 - Object:
 - Instance of classes represented as noun, on which we are doing some actions or operations by using class templates.
 - Let suppose take TV as a Class
 - Data(State)
 - On/off
 - Channel
 - Volume Level
 - Operation (Function)
 - Turn on/off
 - Change channel
 - Change volume



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Object-Oriented Analysis

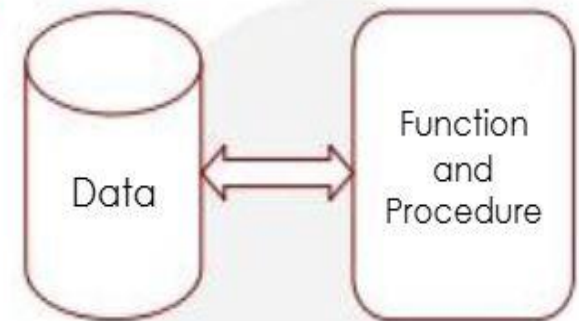
- The main difference between object-oriented analysis and other forms of analysis is that
 - in object-oriented approach, requirements are organized around objects, which integrate both data and functions.
- In traditional analysis methodologies, the two aspects - functions and data - are considered separately.
- The primary tasks in object-oriented analysis are –
 - Identifying objects
 - Organizing the objects by creating object model diagram
 - Defining the internals of the objects, or object attributes
 - Defining the behavior of the objects, i.e., object actions
 - Describing how the objects interact

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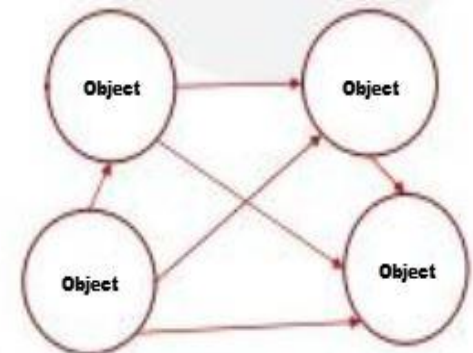
Structured Vs Object Oriented Paradigm

▪ **Structured Paradigm**

- The structured paradigm is a development strategy based on the:-
 - concept that a system should be separated into two parts:
 - Data and functions/procedures .



Structural Paradigm



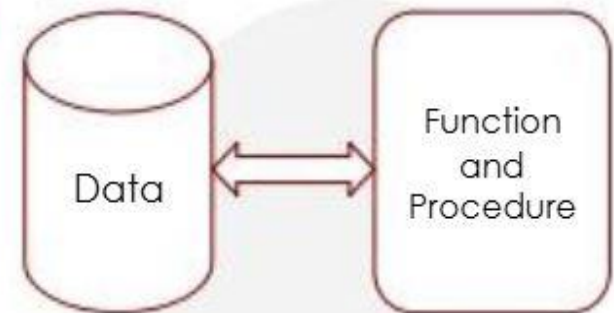
Object oriented Paradigm

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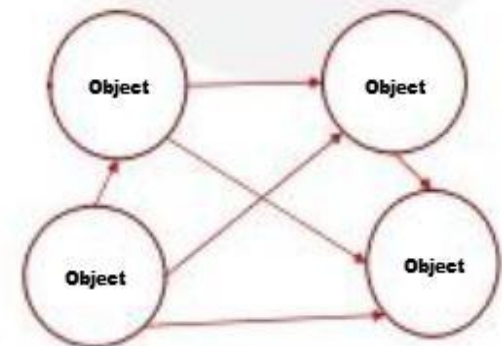
Structured Vs Object Oriented Paradigm

Object Oriented Paradigm

- The main concept behind the object oriented paradigm is that
 - instead of defining systems as two separate parts (data and functions),
 - system is defined as a collection of interacting objects.
- It describes and builds a system that consists of objects.



Structural Paradigm



Object oriented Paradigm

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Benefits of Object Orientation

- **Increased Reusability**

- The object oriented paradigm provides opportunities for reuse through the concepts of *inheritance, polymorphism, encapsulation, coupling and cohesion*.

- **Improved Quality**

- Quality systems are on time, on budget and meet or exceed the expectations of their users.
- Improved quality comes from increased participation of users in systems development.
- So, OO systems development techniques provide greater opportunity for users to participate in the development process.

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Benefits of OO

- **Financial Benefits**
 - Reusability, and improved quality are technical benefits. Object orientation enables you to build systems better, faster and cheaper.
- **Increased Chance of Project Success**
 - A project is successful if it is *on time, on budget and meets the needs of the users*. You need to work closely with users.
- **Reduce Maintenance Burdon**
 - Software organizations currently spend significant resources by maintaining and operating software., it takes significant time to get new projects started.
 - These are problems that object orientation can help you to overcome..

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Drawbacks of OO

- While many benefits exist to OO, Nothing is perfect including object orientation.
 - OO requires greater concentration on requirements analysis and design
 - You cannot build a system that meets users' requirements and Analysis
 - But this fact is often ignored by many developers
 - Developers must closely work with users
 - Users are the experts but they have their own jobs to do (busy)
- OO is just more than programming

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Object Standards

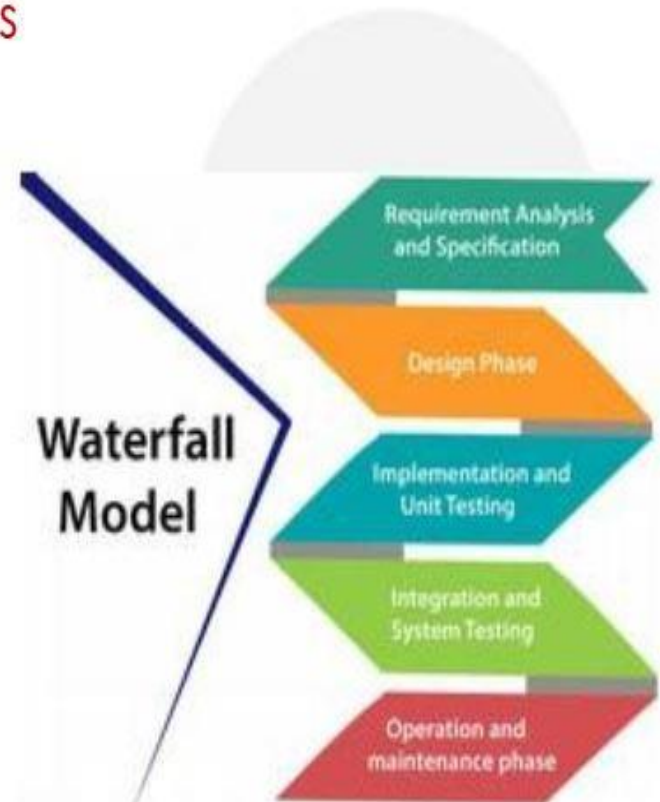
- Basic standards of object for analysis include:
 - Identifying objects
 - Identifying the object relationships
 - Identifying the attributes
 - Identifying services
 - Defining object life cycles

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Software Development Process Models

▪ **Waterfall Model**

- This model is named "Waterfall Model", because its diagrammatic representation resembles a cascade of waterfalls.
- This model has five phases:
 - Requirements analysis and specification,
 - Design,
 - Implementation, and unit testing,
 - Integration and system testing, and
 - Operation and maintenance.



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Software Development Process Models

▪ When to use Waterfall Model?

- It is most suited when
 - The requirements are constant and not changed regularly.
 - The projects are short
 - The situation is calm
 - The tools and technology used is consistent and is not changing
 - The resources are well prepared and are available to use.



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Software Development Process Models

- **Waterfall model**

- **Advantages**

- Simple to implement and minimal resources are required
 - Simple requirements and remain unchanged during development.
 - The start and end points for each phase is fixed.
 - The release date for the complete product, as well as its final cost, can be determined before development.



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Software Development Process Models

- **Waterfall model**

- **Disadvantages**

- The risk factor is higher, not suitable for more significant and complex projects.
 - Cannot accept the changes in requirements during development.
 - It becomes hard to go back to other phase.
 - It does not allow identifying the challenges and risks in the earlier phase, so the risk reduction strategy is difficult to prepare.



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Software Development Process Models

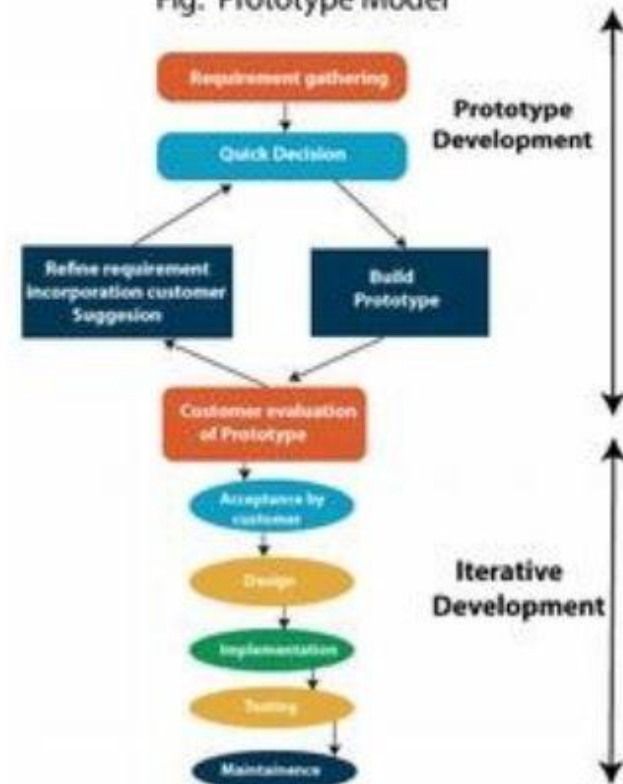
▪ Prototyping Model

- a prototype is built, tested and then reworked as necessary until an acceptable outcome is achieved from which the complete system or product can be developed.

▪ Steps of Prototype Model

- Requirement Gathering and Analysis
- Quick Decision
- Build a Prototype
- User Evaluation
- Prototype Refinement
- Engineer Product

Fig: Prototype Model



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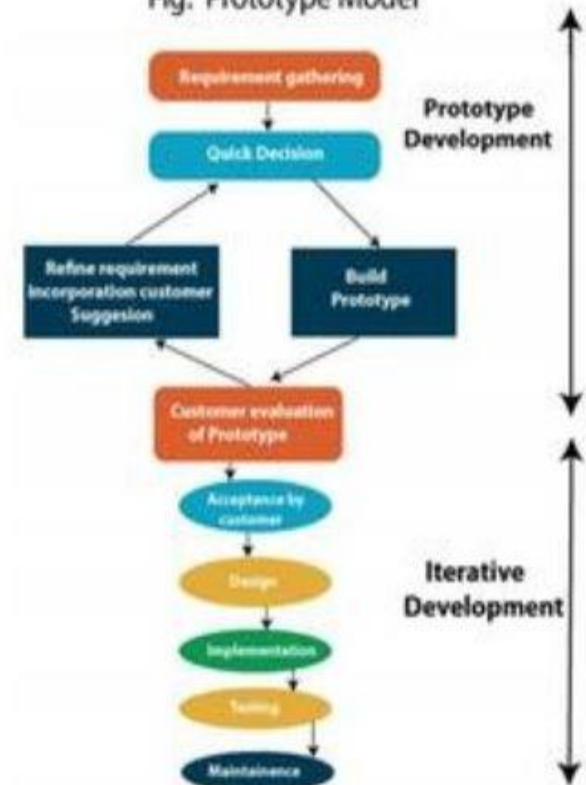
Software Development Process Models

▪ Prototyping Model

▪ Advantage

- Reduce the risk of incorrect user requirement
- Good where requirement are changing/uncommitted
- Reduce Maintenance cost.
- Errors can be detected much earlier as the system is made side by side.

Fig: Prototype Model



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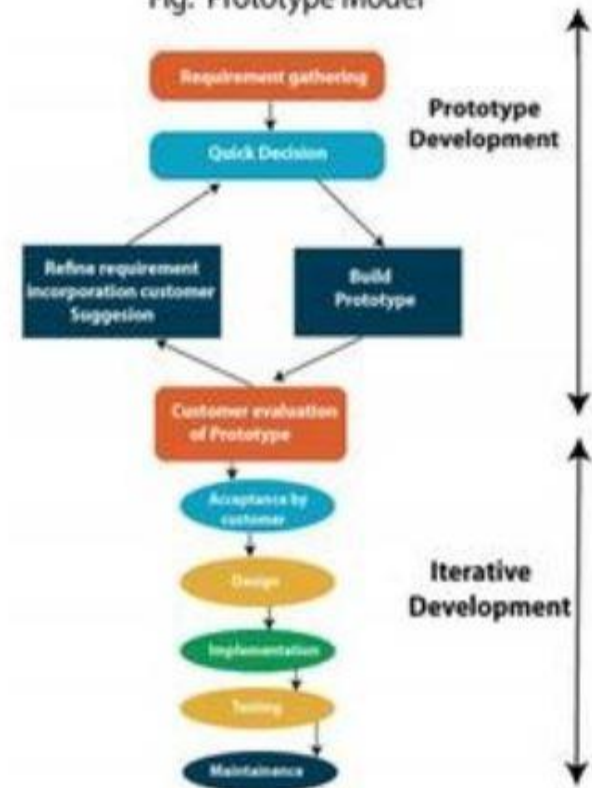
Software Development Process Models

▪ Prototyping Model

▪ Disadvantage

- Require extensive customer collaboration
- Needs committed customer
- Difficult to finish if customer withdraw
- May be too customer specific, no broad market
- Prototyping tools are expensive.
- Special tools & techniques are required to build a prototype.
- It is a time-consuming process.

Fig: Prototype Model

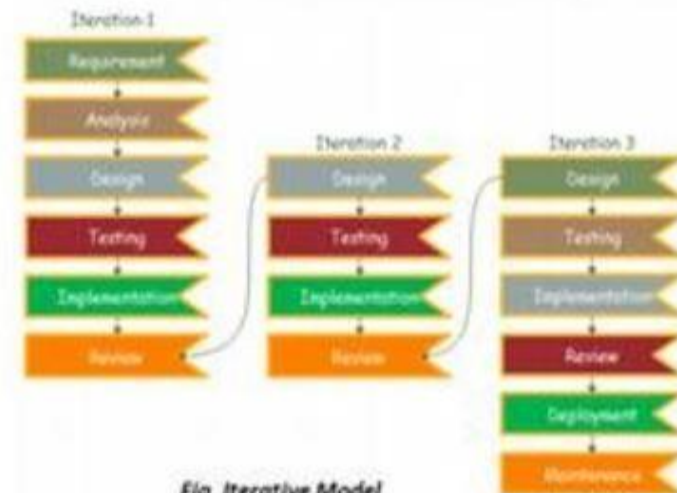


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Software Development Process Models

■ Iterative Model

- The development is in **increment fashion**, each increment adding some functional capability to the system until the full system is implemented.
- At each step, **extensions and design modifications** can be made.
- It is combination of both prototyping and the waterfall model.



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Software Development Process Models

- **Iterative Model**

- Phases

- Requirement gathering & analysis:
 - Design:
 - Implementation:
 - Testing:
 - Deployment:
 - Review:
 - Maintenance:

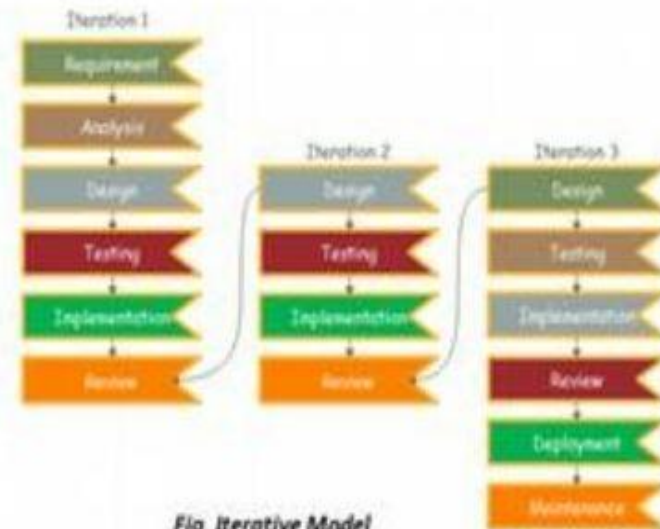


Fig. Iterative Model

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Software Development Process Models

▪ Iterative Model

- When to use the Iterative Model?
 - When requirements are defined clearly and easy to understand.
 - When the software application is large.
 - When there is a requirement of changes in future.
- Advantage
 - Testing and debugging during smaller iteration is easy.
 - A Parallel development can plan.
 - Risks are identified and resolved during iteration.

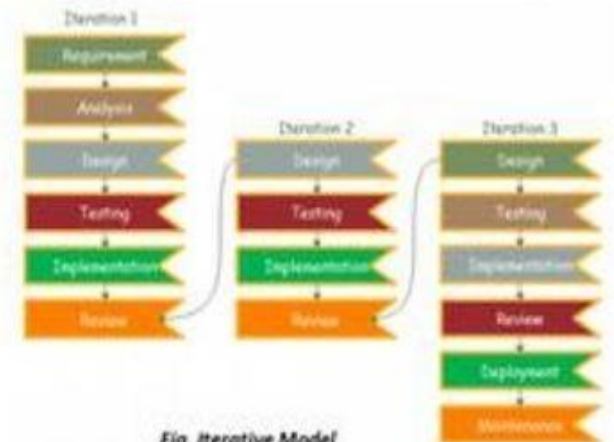


Fig. Iterative Model

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Software Development Process Models

▪ Iterative Model

▪ Disadvantage

- It is not suitable for smaller projects.
- More Resources may be required.
- Design can be changed again and again because of imperfect requirements.
- Requirement changes can cause over budget.
- Project completion date not confirmed because of changing requirements.

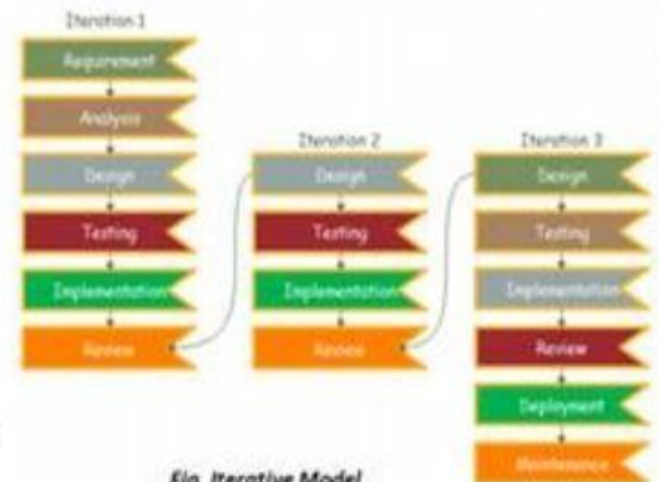


Fig. Iterative Model

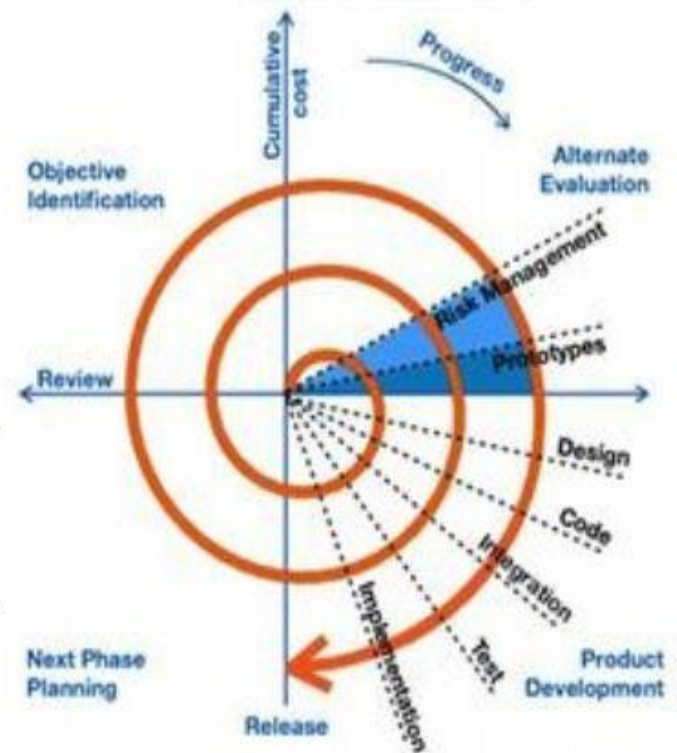
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Software Development Process Models

▪ Spiral Model

- It is combination of both water fall model and Iterative model
- As the name suggests, the activities in this model can be organized like a spiral that has many cycles
- During the early iterations, the additional release may be a prototypes.
- During later iterations, more and more complete versions of the engineered system are produced.

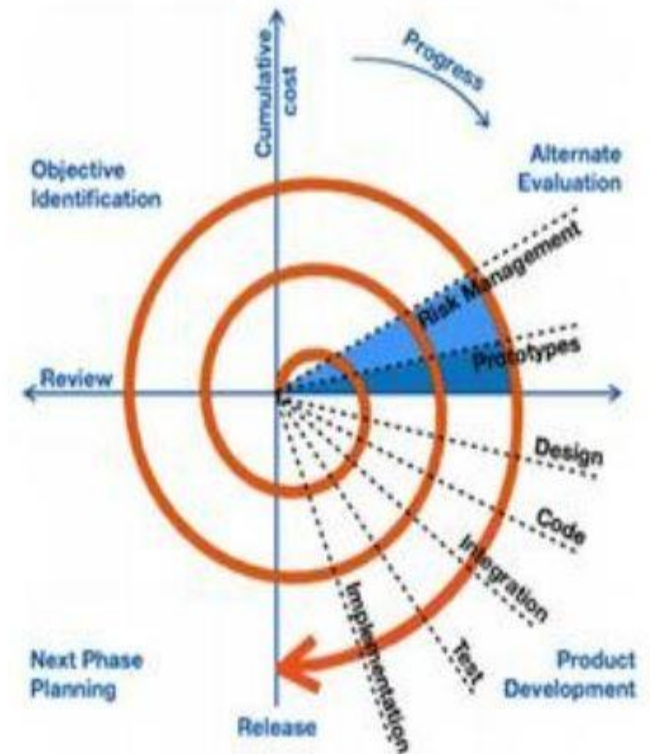


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Software Development Process Models

- **Spiral Model**

- Divided into four parts:
 - Object identification:
 - Alternate Evaluation:
 - Product Development:
 - Next Planning:



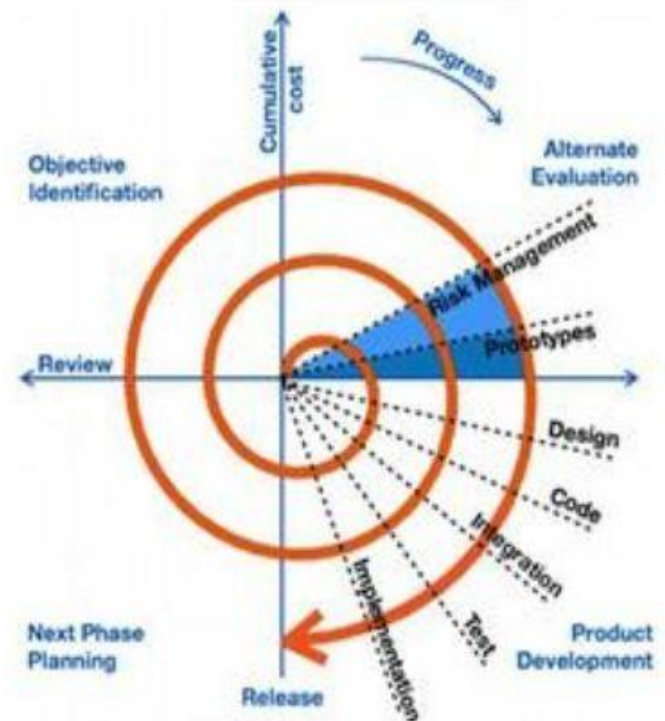
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Software Development Process Models

▪ Spiral Model

- When to use Spiral Model?
 - When deliverance is required to be frequent.
 - When the project is large
 - When requirements are unclear and complex
 - When changes may require at any time
 - Large and high budget projects



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Software Development Process Models

▪ Spiral Model

- Advantages
 - High amount of risk analysis
 - Useful for large and mission-critical projects.
- Disadvantages
 - Can be a costly model to use.
 - Risk analysis needed highly particular expertise
 - Doesn't work well for smaller projects.



- **Thank you!!!!!!!!!!!!**